Microeconomics **Professor Lars Stole**
Business 33801 **Autumn 2020**

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Print your name here.

Final Examination, Business 33801 (AXP21): (Sunday, 3 January 2021)

Read all questions carefully and write answers in the space provided. If you run out of space, you may write on the back of the exam pages; if you run out of space on the back of the exam, you may add additional pages of paper. Graphs must be thoroughly labeled for full credit. Show your work and explain all equations on numeric questions. Be concise – irrelevant remarks will not count for anything. Be careful to budget your time between various parts, taking into account the points allocated to each question (the number in parentheses); these points vary depending upon the question.

There are 15 questions with a total of 180 points possible. You have 180 minutes for the examination, so assign about one minute per point. Allocate your time wisely.

YOU MUST READ AND SIGN THE HONOR STATEMENT BELOW IN ORDER TO RECEIVE CREDIT FOR THE EXAMINATION.

> I pledge my honor that I have not violated the Honor Code during this examination. I will not communicate any information related to this examination to *any* individual that has not already seen the examination <u>until after 7 January 2021</u>.

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Sign your name here.

True-False-Uncertain (48 points total)

Indicate whether the italicized portion of each statement is True, False, or Uncertain and **EXPLAIN your answer**. You may take the non-italicized portion of the statement as true. No credit will be given for answers without explanation. *Read each question carefully.*

Question 1 (**6 points**) **True, False, Uncertain? [Explain your answer.]** When the price of whisky is p_w and the price of ale is p_a, Gamora purchases some whisky ($w > 0$) but no ale ($a = 0$) . *For Gamora, at her optimal choice of beverages,*

$$\frac{MU_a}{MU_w} > \frac{p_a}{p_w}.$$

Question 2 (**6 points**) **True, False, Uncertain? [Explain your answer.]** The elasticity of demand for wheat is $\varepsilon^d = -0.5$, and this elasticity is constant. The supply of wheat is infinitely elastic. The current equilibrium price is $p^* = 5$ per bushel. The government proposes a new subsidy to wheat farmers equal to $s = 0.50$ per bushel. *The subsidy will increase the equilibrium consumption of wheat by approximately 5 percent.*

Question 3 (**6 points**) True, False, Uncertain? [Explain your answer.] The market supply curve for milk is infinitely elastic at a price of \$2 per gallon. The market demand curve for milk is downward sloping. The government introduces a tax of \$0.10 per gallon of milk and the quantity of milk sold falls by 1%. *Demand for milk is inelastic.*

Question 4 (**6 points**) **True, False, Uncertain? [Explain your answer.]** Building office towers in Nova city is a competitive industry. The long-run supply curve for office space in the city is upward sloping. *The builders of office towers in Nova city make economic profit.*

Question 5 (**6 points**) True, False, Uncertain? [Explain your answer.] *If a monopolist raises its price slightly above its profit-maximizing price, it will reduce its revenue.*

Question 6 (**6 points**) **True, False, Uncertain? [Explain your answer.]** Yondu is a monopolist selling a unique kind of dashboard trinket in a small town. The per-period demand for these trinkets is

$$q^d = 200 - 2p.$$

Yondu maximizes profits by selling $q^* = 80$ trinkets each period. *Yondu's marginal cost is greater than 25.*

Question 7 (**6 points**) **True, False, Uncertain? [Explain your answer.]** The market for anulax batteries is perfectly competitive. Because of a change in technology, the supply curve for these batteries shifts outward. *The market expenditures on anulax batteries will increase.*

Question 8 (**6 points**) **True, False, Uncertain? [Explain your answer.]** Consider a competitive market for neodymium (Nd is an elemental rare-earth metal). Neodymium is used in making magnets by U.S. firms and by foreign firms. There are 100 US magnet manufacturers who demand Nd; each has an elasticity of demand equal to $\varepsilon^d_{us} = -0.75$. There are 200 foreign magnet producers who demand Nd; each has an elasticity of demand equal to $\varepsilon^d_f = -1.5$. No one else demands Nd other than these 300 firms. The US firms currently buy half of all Nd that is sold. *The <u>market</u> elasticity of demand for Nd is -1.25.*

Short questions (132 points)

Question 9 (30 points) TigerKing (TK) industries produces ExoticBalm, a unique form of tiger balm (a medication to alleviate pain), and is currently a monopoly because no one can figure out the secret ingredient in ExoticBalm (and hence no one can make a competing product in this market). TK can produce up to 25 cases of balm at a constant unit (opportunity) cost of $MC = AC = 2$ per case (up to capacity). The demand curve for ExoticBalm is

$$p = 100 - Q.$$

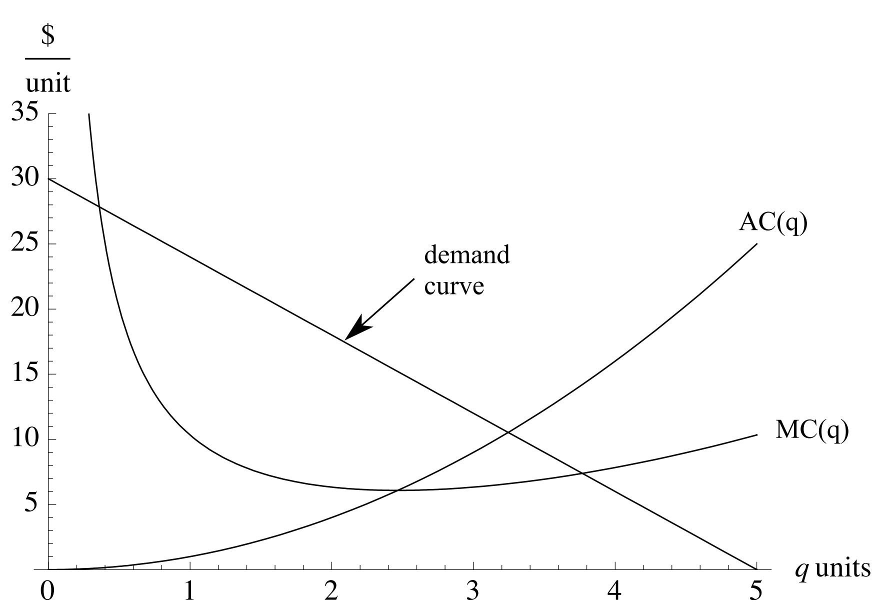
(a). How many cases of ExoticBalm should TK sell in each period and at what price? Note that TK cannot produce more than 25 units. Explain.

(b). What is the elasticity of demand at the optimal price found in part (a)?

(c). Carole discovers the secret ingredient in ExoticBalm and figures out a way to produce a perfect substitute to what TK sells. Carole has a constant unit cost of $MC = AC = 5$ per unit produced, but she has unlimited capacity at this price. If Carole thinks TK will produce at capacity, $q_{tk} = 25$, how much should Carole supply to the market? Explain. [Hint: This problem is equivalent to a monopolist facing a competitive fringe supplying 25 units independent of market price.]

(d). Continuing with the facts in (c), is the optimal output you computed for Carole in (c) a Nash equilibrium? That is, is it optimal for TK to produce at capacity, $q^*_{tk} = 25$, providing that Carole produces according to your answer in (c)? Explain.

Question 10 (6 points) Find the major error in the following picture. Briefly explain.



Question 11 (22 points) Quill owns a local hardware (DIY) store in Chicago which sells saws to two types of potential buyers, rich and poor. Quill cannot distinguish between the two types, but knows that the rich buyers value the store's saw at \$52 while the poor buyers value the saw at \$30. The marginal and average opportunity cost of the saw to the store is \$10. The two types of consumers are equally likely (probability 1/2 of each type).

(a). If Quill must set a single price for the saw, what price maximizes his expected profit? Explain.

(b). Quill is now considering the use of rebates. A rebate is a coupon which any buyer can mail in with a sales receipt to claim a refund of a fixed amount from the store, usually paid several weeks later. Going through this process is costly to the buyers. Suppose that the personal cost of sending in a rebate claim for a rich buyer is \$14 and the personal cost of sending in a rebate claim for the poor buyer is only \$4. As an example, a \$20 rebate would effectively be worth only \$6 to a rich buyer, but it would be worth \$16 to a poor buyer.

Suppose that the rebate processing costs of the store are sufficiently small that they can be ignored. Should Quill offer his customers a rebate? If so, what is the optimal price of the saw and what is the optimal value of the rebate? If not, why not? Explain.

Question 12 (24 points) The market for railroad freight service is perfectly competitive with an industry capacity of 100 units of freight. The average and marginal opportunity cost of each unit is $MC = AC = 35$ for every railroad company in the industry, up to capacity. There are two customer segments – farmers who ship grain and oil companies that ship oil. Each segment has the same demand curve for freight service, given by

$$p_f = 120 - q_f^d,$$

for the farmers and

$$p_o = 120 - q_o^d$$

for the oil companies.

(a). What is the equilibrium price in this competitive market and how much freight is sold in each segment? Explain.

(b). Suppose now that the government decides to set a maximum price for freight in the farmers' market to $\overline{p}_f = 40$, but allows the market to determine the price for oil freight. Furthermore, assume that the railroads must supply all of the freight demanded by farmers before they are allowed to serve any demand by the oil companies. (E.g., farmers always get in line ahead of the oil companies.) What is the equilibrium price of freight for oil companies? Explain.

(c). Under the government regulation described in (b), what is the nature of the deadweight loss that arises from the government price control? [Note: You should provide the economic "story" or description of the source of deadweight loss rather than a graph or a calculation.]

Question 13 (16 points) The price of corn syrup in the U.S. has recently increased by 20 percent because of the increase in the price of corn. (Corn syrup is a sweetener used in making foods.) Following the increase in the price of corn syrup, the equilibrium quantity sold of raw sugar (a demand substitute for corn syrup) increases by 10 percent.

(a). Draw supply and demand graphs for the corn syrup market and for the sugar market to illustrate how the change in the price of corn (an input to corn syrup) affects the two markets.

(b). Suppose that the own-price elasticity of demand for sugar is constant and equal to -10 and that the supply elasticity for sugar is constant and equal to 5. If only the prices of sugar and corn syrup change, and if the supply curve of sugar is unaffected by the price of corn, what is the percentage price increase in sugar?

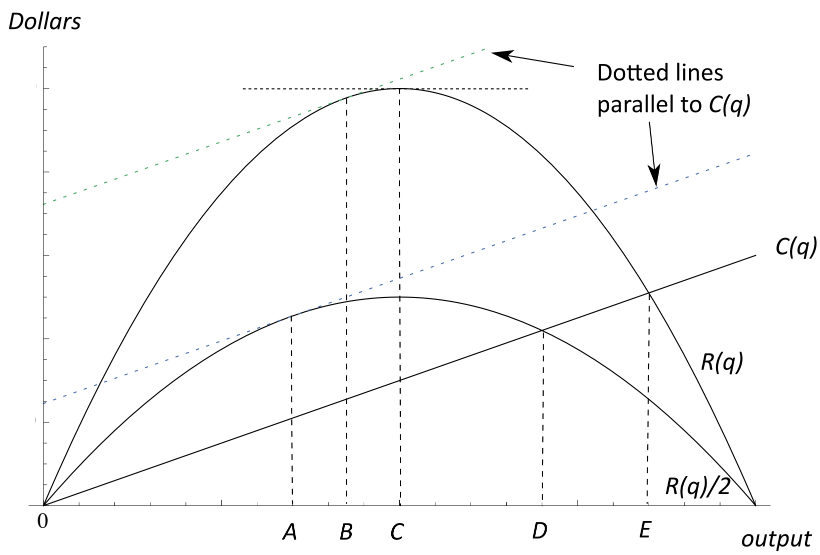
[Hint: remember that the changes in price and quantity given above are equilibrium outcomes. Supply must equal demand both before the increase in corn prices and after the increase.]

Question 14 (16 points) Nebula owns a company that makes pre-packaged sandwiches for convenience stores. Nebula is a price-taker. Her daily opportunity cost for making sandwiches is $C^O(q) = \frac{1}{2}q^2 + 8$ for $q > 0$ (and $C^O(0) = 0$). Her marginal cost is $MC(q) = q$.

(a). At what market price is Nebula indifferent between operating and shutting down? Explain.

(b). If the market price for a sandwich is \$10, how many sandwiches should Nebula produce each day to maximize profits?

Question 15 (18 points) A book publisher has a contract which pays 1/2 of all book revenues to the author. The printing and distribution costs, however, are entirely paid for by the publisher. In the figure below, the publisher's total opportunity cost function, $C(q)$, the total revenue function for books, $R(q)$, and 1/2 of the total revenue function for books are plotted together.



(a). How many copies of the book does the publisher wish to sell? (Choose among A, B, C, D and E.) Explain.

(b). How many copies of the book does the author wish to sell? (Choose among A, B, C, D and E.) Explain.

(c). Who will prefer to sell at a lower price: the author or the publisher? Explain.

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Suggested answers to exam questions

1 False. One way to explain this is to note that if whisky is purchased, but ale is not, then the "rate of return" on whisky must exceed that of ale:

$$\frac{MU_w}{p_w} \geq \frac{MU_a}{p_a}.$$

But this implies that

$$\frac{p_a}{p_w} \geq \frac{MU_a}{MU_w},$$

a contradiction with the facts.

2 True. Because the wheat supply is perfectly elastic, the supply curve is horizontal. Thus, any subsidy will be passed along to the consumers. This implies that price will decrease from $p_0^* = 5$ to $p_1^* = 5 - 0.50 = 4.50$. This is an (approximately) 10 percent reduction in price. Using the elasticity of demand equal to -0.5, this would translate into an increase in consumption equal to $(-10)(-0.5) = 5$ percent.

3 True. Because supply is perfectly elastic (i.e., the supply curve is horizontal), the entire tax gets passed along to the buyers and the market price rises from \$2 to \$2.10 (about 5 percent). The corresponding decrease in demand is only 1 percent. Hence the elasticity of demand is $\varepsilon^d = \frac{-1\%}{5\%} = -0.20$, which is inelastic.

4 False/Uncertain. Builders may be homogenous and plentiful, therefore earning no economic profit. The fact that the long-run supply curve is upward sloping only implies that someone in the vertical production chain is earning economic profit. For example, the economic profit might entirely go to the owners of the land in Nova city.

5 True. If $MC \geq 0$ and the monopolist is maximizing profits, then the firm will choose to price in the elastic region ($|\varepsilon| \geq 1$) and an increase in price (and a corresponding decrease in units sold) will always decrease revenue.

6 False. Yondu's inverse demand curve is $p = 100 - \frac{1}{2}q$, and thus his marginal revenue at $q = 80$ is $100 - 80 = 20$. If his marginal cost were 25 or higher, his marginal revenue would be less than marginal cost and he would not be maximizing profits.

7 Uncertain. The shift in the supply curve causes the equilibrium point to move down the demand curve. If demand is elastic in this region, expenditures will increase; if demand is inelastic, expenditures will decrease.

8 False. A one percent increase in price leads the US demand for Nd to fall by 0.75 percent and it leads to foreign demand to fall by 1.5 percent. Because the US buys half the world supply of Nd, aggregate admen falls by

$$\frac{1}{2}0.75 + \frac{1}{2}1.5 = \frac{9}{8} = 1.125$$

percent. The number of firms in the US and abroad is irrelevant to determining the elasticity of demand.

9 (a). First consider TK's marginal revenue function:

$$MR(q_{tk}) = 100 - 2q_{tk}.$$

At a marginal cost of 2, TK would like to produce

$$100 - 2q_{tk} = 2,$$

or $q_{tk} = 49$ units. Joe can only supply at capacity of 25, so the optimal amount to supply is $q^*_{tk} = 25$. At this output, $MR_{tk} = 50$ which greatly exceeds marginal cost, but he can't produce any more. Selling just $q^*_{tk} = 25$ units allows TK to charge $p^* = 100 - 25 = 75$ per unit.

(b). The formula to use is

$$\varepsilon^d = \frac{dQ^d}{dp}\frac{p^*}{Q^*} = (-1)\frac{75}{25} = -3.$$

(c). Suppose that $q_{tk} = 25$. Carole's residual demand curve is given by

$$p = 100 - (q_c + 25) = 75 - q_c.$$

With this demand, Carole's marginal revenue is

$$MR_c(q_c) = 75 - 2q_c.$$

Setting this equal to $MC_c = 5$,

$$75 - 2q_c = 5,$$

so $q^*_c = 35$. Given $Q^* = 25 + 35 = 60$, the market price is $p^* = 100 - 60 = 40$.

(d). Given that Carole is choosing $q^*_c = 35$, we can compute TK's residual demand curve:

$$p = 100 - (35 + q_{tk}) = 65 - q_{tk}.$$

Thus, TK's marginal revenue is

$$MR_{tk}(q) = 65 - 2q_{tk}.$$

Note that at $q_{tk} = 25$, this marginal revenue is 15, which exceeds marginal cost. Hence, TK can do no better than produce at capacity. $q^*_{tk} = 25$ and $q^*_c = 35$ is a Nash equilibrium to the Cournot duopoly game.

10 The error in the graph is to have the AC curve going through the minimum of the MC curve. Their roles should be switched in this regard.

11 (a). There are two choices. Set a price at $p = 52$ and sell to only half of the potential consumers. Profit per consumer would be $\frac{1}{2}(52 - 10) = 21$. Alternatively, set a price of $p = 30$ and sell to all buyers. Profit per consumer would be $(30 - 10) = 20$. Hence, it is optimal to set the price at $p^* = 52$ and sell to only half of your potential buyers.

(b). The data of the problem (along with the rebate costs) are given as follows:

	Rich buyers	Poor buyers
Value of saw	52	30
Rebate-processing cost	14	4
Value of saw less rebate cost	52-14=38	30-4=26

We can rearrange this data by denoting the combined value of the saw and rebate costs (the base good) and then the value of the saw when no rebate costs needs to be expended (the "upgrade"):

Quality	Rich buyers	Poor buyers
Value of saw with rebate cost (base good)	52-14=38	30-4=26
Value of removing rebate cost (upgrade)	14	4

Now we can apply our analysis from class using this table and the fact that the saw costs 10 to produce. Consider the marginal price for the saw with rebate (the first row above). If the seller offers a price of $p_1 = 38$, then only the rich buyers will purchase it and he will make an average profit of $\frac{1}{2}(38 - 10) = 14$ per buyer. If, instead he sold it at $p_1 = 26$, then both buyers would purchase the base good and average profit per buyer would be $26 - 10 = 16$. Hence, it is better to sell to both consumer types and set $p_1^* = 26$. Now consider the "upgrade" (the second row): at a price of $p_2 = 14$, only the rich buyer would purchase the upgrade, so the firm would make $\frac{1}{2}(14) = 7$. If, instead the merchant sold the upgrade at $p = 4$, both types would pay to avoid the rebate and the firm would make 4. Thus, $p_2^* = 14$ is optimal. Together, we have that the optimal price for the saw (without rebate) is $p_{saw}^* = p_1^* + p_2^* = 26 + 14 = 40$ and the value of the rebate is $r^* = 14$.

12 (a). First, note that the aggregate demand function for freight is

$$Q^d = q_o^d + q_f^d = 240 - 2p.$$

There are two possible outcomes to consider for the equilibrium – industry supplies at capacity or the industry supplies less than capacity. If the industry supplies at capacity, then we must have

$$Q^d = 100 \implies 240 - 2p = 100,$$

so $p^* = 70$. Notice that at this price, $p^* > MC$, so firms would like to produce more but they can't because they are at capacity, so we have determined the equilibrium price. [Note: If you considered that firms might be pricing at $p^* = MC = 35$, then $Q^d = 240 - 2(35) = 170$, but this exceeds capacity so it must be that $Q^d = 100$ and the first case arises.] Thus, we have $p^* = 70$ and $Q^* = 100$. At this price, each segment will consume 50 units.

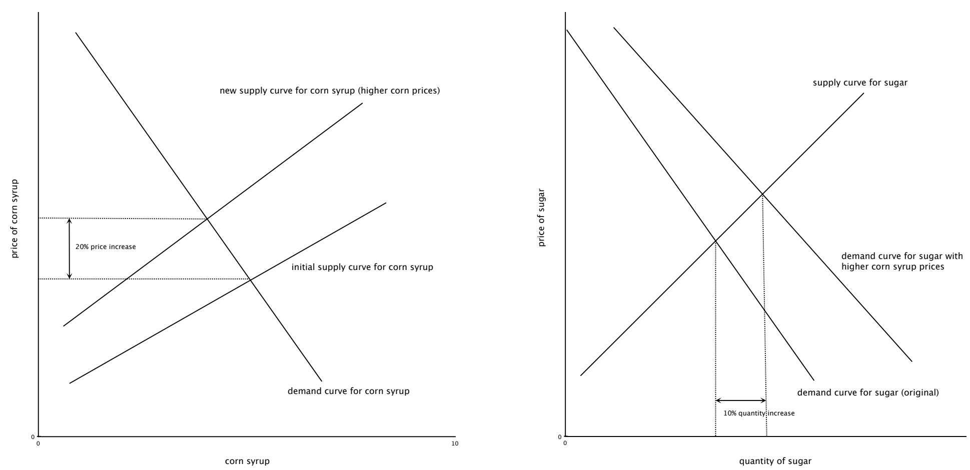
(b). Given that $p_f = 40$ and the market must supply $q_f = 120 - 40 = 80$ units of freight to the farming segment, only 20 units of freight remain for the oil industry. The equilibrium price in the oil segment must equate demand with supply, and thus

$$q^d = 120 - p_o = 20 \implies p_o^* = 100.$$

(c). Regardless of the government price control or not, the industry supplies 100 units of freight. The only deadweight loss that arises is generated by the wrong consumers buying the freight. The marginal farmer values the freight at only 40 while the marginal oil company values it at 100. There are unrealized gains from trade between these two groups.

[Aside: if you actually calculated the consumer and producer surpluses under the price control setting in (b), you would find that $CS_f = \frac{1}{2}(120-40)(80) = 3200$, $CS_o = \frac{1}{2}(120-100)(20) = 200$ and $\Pi_{ind} = (40-35)(80) + (100-35)(20) = 400 + 1300 = 1700$. Thus, $W = CS_f + CS_o + \Pi_{Ind} = 5100$. Compare this to the total surplus under (a) which is $W = 2500 + 3500 = 6000$. The DWL is therefore equal to 900. Of course, I did not ask for this computation, only the economic story.]

13 (a).



(b). We know that the quantity of sugar sold increases by 10 percent. If the supply curve for sugar is stable with an elasticity of 5, then we know we must have a price increase of 2 percent to generate this increase.

14 (a). Nebula's minimum AC arises where $MC = AC$. Thus, the output at this intersection, $\hat{q}$, satisfies

$$AC(\hat{q}) = MC(\hat{q}) \iff \frac{\frac{1}{2}\hat{q}^2 + 8}{\hat{q}} = \hat{q}.$$

Using some algebra, we can multiple each side by $\hat{q}$ to obtain

$$\frac{1}{2}\hat{q}^2 + 8 = \hat{q}^2,$$

or simplifying further

$$\hat{q}^2 = 16.$$

Thus, $\hat{q} = 4$. The associated minimum AC is found by plugging $\hat{q} = 4$ into the average cost function: $AC(\hat{q}) = \frac{1}{2}(\hat{q}) + \frac{8}{\hat{q}} = 4$. This is the critical price to determine supply. At any price higher, the firm can make profits because there are some outputs for which price exceeds average cost. Below this value, price is less than average cost for every output.

(b). The marginal output rule requires that $MR = MC$ if production is positive. Given Nebula is a price taker and the price is $p = 10$, we have $MR = 10$. Thus, the marginal output rule requires $10 = MC = q$, so $q^* = 10$ sandwiches per day – assuming it is optimal to operate. We still need to confirm that Nebula should operate rather than shutdown. Given the price is above the price found in (a) above, i.e. $p = 10 > \min AC = 4$, she should operate. Alternatively, we can computer her revenue at $q^* = 10$ which is $R = 10(10) = 100$; her opportunity cost is $C^O(10) = \frac{1}{2}(100) + 8 = 58 < R = 100$. Thus, economic profit is positive and Nebula should operate with $q^* = 10$.

15 (a). The publisher will choose the output for which the publisher's marginal revenue equals marginal cost. Hence, $\frac{1}{2}R'(q) = MC(q)$, so the slopes of the total cost and the publisher's revenue function, $\frac{1}{2}R(q)$, are the same. This is true only at point A.

(b). The author will prefer to maximize residual revenues which is the same as maximizing revenues. Thus, he would want the peak of the revenue function. This occurs at point C.

(c). The author will want to sell more than the publisher because the author does not bear any of the marginal costs of production and distribution. That is why C is higher than A. In terms of prices, this means that the author prefers a lower price than the publisher.